

# KAĞAN ONUR

Data Analyst • Data Scientist

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## PROFILE

Statistics graduate of Yıldız Technical University (2026). What interests me is getting from data to a decision: my thesis modelled 21 years of Turkish investment instruments to measure which one actually protects savings against inflation and currency risk; a panel data project tested whether discount campaigns genuinely change buying behaviour; and I represented Türkiye at ASA Data Quest 2025 alongside teams from four continents. I work in Python, R, Stata and SQL, and I treat explaining *why* a model behaved as it did as part of the job rather than an afterthought. Alongside this I designed, built and shipped mesite.org single-handedly. Open to data analyst and data scientist roles.

## SELECTED PROJECTS

### Time Series Analysis of Turkish Investment Instruments

Undergraduate Thesis • Yıldız Technical University • 2026

Advisor: Prof. Dr. Öyküm Esra Yiğit

I set out to answer two questions: which asset actually protects savings against Türkiye's high inflation and currency volatility, and whether a deep learning model can forecast these markets better than a classical one. I used 21 years of monthly data (2005–2025) on spot gold, USD/TRY, CPI inflation, deposit rates and the BIST 100.

- Gold showed evidence of a long-run hedge against both inflation and currency risk — over two decades it was the one instrument that preserved real purchasing power while deposit returns eroded.
- Accounting for crises changed the picture substantially: once the 2008, 2018 and 2021 structural breaks were built into the model, its explanatory power for gold rose fourfold ( $R^2$  0.12 → 0.51).
- Contrary to expectation, classical ARIMA beat the deep learning LSTM ( $R^2$  0.988 vs 0.865 on USD/TRY). I traced the cause rather than stopping at the number: the comparison was less fair than it looked, because the LSTM forecasts on top of its own forecasts and accumulates error across 46 months.
- Currency moves drive inflation, not the reverse ( $p < 0.001$ ) — a quantified measure of exchange-rate pass-through.

Methods: ADF / PP / KPSS unit-root tests • Bai–Perron structural breaks • ARDL bounds testing • ARMA-GARCH(1,1) and EGARCH volatility models • Granger causality • ARIMA/SARIMA vs LSTM vs Holt benchmarking. Python (statsmodels, arch, TensorFlow/Keras).

### MeSite — Branching Video Platform

Individual project • mesite.org • 2026, ongoing

A video platform where stories branch: you watch a clip, and if you don't like where it goes you upload your own continuation. I designed, built and deployed the whole system — data model, ranking algorithm, API, interface and server. It is live in closed beta.

- Designed the ranking algorithm that decides which branch becomes the main storyline. Retention is measured in **capped seconds rather than completion rate** — completion rate would hand a perfect score to any 8-second clip, while raw watch time would simply reward length; capping neutralises both.
- Made diversity and retention **multipliers rather than additive terms**, so a branch inflated by a single author or watched by nobody cannot become canon no matter how large it grows — one author scores 0.50 against 0.67 for two.
- Built the platform on two databases: **PostgreSQL** for accounts and transactions, **Neo4j** for the branching graph, since ancestry and descendant counts are the core query and a relational schema would have made them expensive.
- Instrumented watch-time collection specifically so the algorithm's coefficients can be calibrated against real behaviour once the beta has enough data, rather than staying at their current estimated values.
- Shipped it end to end: Docker deployment, automated nightly backups with a tested restore path, rate limiting, secret rotation, and account deletion with pseudonymisation to meet Turkish data protection law (KVKK).

Stack: Python (FastAPI, SQLAlchemy), PostgreSQL, Neo4j, Docker & Docker Compose, Caddy, JWT authentication, FFmpeg video transcoding, JS/HTML/CSS front end. ~8,600 lines of Python across 32 modules with an automated test suite. Deployed on a Hetzner VPS.

### Effect of Price Discounts on Player Counts on Steam

Panel Data Analysis • Team of 3 • 2026

Do Steam sales actually bring players back? We tested the effect of the platform's four large seasonal sales on player numbers, using twelve months of data on nine popular games.

- Price genuinely matters: a **1% price increase reduces player count by 0.57%** — the law of demand holds on a digital platform where marginal cost is zero.
- But the sales themselves did not move the needle. None of the summer, winter, spring or autumn sale periods produced a statistically significant lift in players.

- The reason turned out to be a measurement trap: **98% of the variation in player numbers is explained by how big the game already is**. Discounts don't grow a game — large games simply stay crowded. Anyone who didn't separate differences *\*between\** games from changes *\*within\** a game would have concluded that sales work.
- I combined discount depth and discount duration into a single "discount pressure" measure, which removed collinearity between the two and made the coefficient directly interpretable.

Methods: balanced panel of 108 observations (9 games × 12 months) · Levin-Lin-Chu, Im-Pesaran-Shin and Fisher-ADF panel unit-root tests · Hausman test selecting fixed effects ( $\chi^2 = 9.27$ ,  $p = 0.026$ ) · game-clustered robust standard errors. Stata, R.

Data Science & Sustainable Development

ASA Data Quest 2025 · International · Representing Türkiye

American Statistical Association (CIRS) · Mentor: Ceylan Yozgatligil · Team of 4

We asked whether countries have to trade environmental cost for development, and presented Türkiye's entry against teams from Brazil, Indonesia, India, Kenya and China.

- Wealth, education and life expectancy move together very tightly (correlations of **0.87–0.92**) — economic capacity and human development reinforce each other globally.
- Environmental cost splits along a clear line: national income correlates **0.87** with CO<sub>2</sub> emissions while education and health spending correlate weakly or negatively — social investment buys development at a far lower carbon price.
- A model optimising all four dimensions together — economy, education, health and emissions — identified the Nordic countries and Costa Rica as the most balanced performers, showing the trade-off is not inevitable.

Validation: Durbin–Watson, VIF and Shapiro–Wilk diagnostics.

Customer Purchasing Behaviour Analysis

Team of 3 · 2026

We examined what drives a customer's decision to buy, testing the effect of age, gender and review quality on an open-access dataset. Gender emerged as a statistically significant driver, while the other factors did not hold up under testing.

Methods: multivariate logistic regression · Pearson chi-square tests.

EXPERIENCE

Data Annotation Specialist

Jan 2026 – Feb 2026

Artiwise — AI-Powered Software Company

- Labelled and classified text data for supervised NLP models at a company building AI-powered document analysis for customer-centric process management.
- Applied and refined annotation guidelines to keep labels consistent, and ran quality checks on completed batches to keep training data reliable.

EDUCATION

Yıldız Technical University

2022 – 2026

BSc Statistics, Faculty of Arts and Sciences

Relevant coursework: Econometrics, Time Series Analysis, Panel Data Analysis, Multivariate Statistics, Regression Analysis, Data Visualization, Data Management

İstanbul Kültür University

2020 – 2022

Computer Engineering, Faculty of Engineering

English preparatory programme completed; transferred to Statistics at YTU

Gürlek Nakipoğlu Anatolian High School

2015 – 2019

TECHNICAL SKILLS

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Programming & Tools        | Python (pandas, NumPy, statsmodels, scikit-learn, TensorFlow/Keras) · R · Stata · SQL · Excel · SPSS                    |
| Time Series & Econometrics | ARIMA / SARIMA · ARMA-GARCH · EGARCH · ARDL bounds testing · Granger causality · Bai–Perron · ADF / PP / KPSS           |
| Panel & Regression         | Fixed and random effects · Hausman test · cluster-robust standard errors · panel unit roots · OLS · logistic regression |
| Databases & Engineering    | PostgreSQL · Neo4j (graph database) · FastAPI · Docker · Git · REST API design                                          |
| Machine Learning           | LSTM and deep sequence models · supervised classification · model benchmarking (MAE, MAPE, R <sup>2</sup> )             |
| Data Work                  | Cleaning and merging multi-source data · feature engineering · data labelling · visualization and reporting             |
| Languages                  | Turkish — Native · English — Upper-Intermediate (B2), DİLKO in partnership with Pearson (2022)                          |